

Likelihood-free estimation by matching random features

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- ▶ **Temporal dependence:** X_t^{obs} may depend on the past $X_{t-1}^{\text{obs}}, X_{t-2}^{\text{obs}}, \dots$
- ▶ **Nonstationarity:** Joint distribution of $X_{t_1+\tau}^{\text{obs}}, \dots, X_{t_k+\tau}^{\text{obs}}$ may change with τ

Simulation-based estimation

- For some $\theta_0 \in \Theta \subset \mathbb{R}^p$, assume the time series is *generated by nature* as

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- ▶ **Key assumption:** Can simulate from an *approximation* $G_t^{(i)}$ to G_t at any $\theta \in \Theta$
 - 1 Approximation improves as $i \rightarrow \infty$
 - 2 Growing burn-in periods, discretized ODEs, etc.

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 - ▶ Extract early layer of neural network

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 - ① Φ is **one-to-one** on compact subsets of $\mathbb{R}^p$
 - ② Φ has a **nice derivative** on certain compact subsets of $\mathbb{R}^p$
 - Jacobian of Φ has full rank on compact subsets of smooth manifolds in $\mathbb{R}^p$

What do we mean by “almost every” function? Prevalent subset

Definition (J. Yorke and Ott (2005))

Let V be a completely metrizable topological vector space. A Borel set $E \subset V$ is said to be **prevalent** if there exists some Borel measure μ on V such that:

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- ▶ If $E \subset V$ contains a prevalent Borel set, then “almost every” element of V is in E
 - ▶ In this sense, “almost every” C^1 smooth map from $\mathbb{R}^p$ to $\mathbb{R}^{2p+1}$ has the properties from the embedding theorem of Sauer, J. A. Yorke, and Casdagli (1991)

Main ideas for estimation via random features

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- ▶ Under smoothness conditions on $\theta \mapsto P_\theta$, the expectations of $2p+1$ “generic” continuous, bounded functions $\varphi : \mathbb{R}^{(m+1) \times d} \rightarrow \mathbb{R}^{2p+1}$ will be C^1 smooth in θ

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- ▶ For some φ , define the C^1 smooth function $\Phi : \mathbb{R}^p \rightarrow \mathbb{R}^{2p+1}$ by extending the map

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 - That is, Φ will be one-to-one on Θ and have a nice derivative on certain subsets
- ▶ **Key idea:** Estimate θ_0 by *minimizing distance* of $\hat{\Phi}(\theta_0)$ and $\hat{\Phi}(\theta)$ over $\theta \in \Theta$,
 $\hat{\Phi}(\theta)$ is a time-average of $2p + 1$ random features of a time series generated at θ

Random Fourier features

- ▶ Let $x_1, \dots, x_{m+1}$ be vectors in $\mathbb{R}^d$, and define $x = (x_1, \dots, x_{m+1})^\top \in \mathbb{R}^{(m+1) \times d}$

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- ▶ We consider random features of the form

$$\varphi_i(x) = \cos \left(\sum_{j=1}^{m+1} \Omega_{i,j} \cdot x_j + \alpha_i \right)$$

where $\Omega_{i,j} \stackrel{\text{iid}}{\sim} N(0, I_d)$ and $\alpha_i \stackrel{\text{iid}}{\sim} U(-\pi, \pi)$ for $i = 1, \dots, k$ and $j = 1, \dots, m+1$

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- ▶ Denote all k of these functions by

$$\varphi = (\varphi_1, \dots, \varphi_k)$$

so that $\varphi : \mathbb{R}^{(m+1) \times d} \rightarrow \mathbb{R}^k$

Random Fourier features

- ▶ Define the i -th random feature at time $t = m + 1, \dots, n$ as

$$f_{t,i}^{\text{obs}} = \varphi_i(X_{t-m:t}^{\text{obs}}), \quad f_{t,i}^{(r)}(\theta) = \varphi_i\left(X_{t-m:t}^{(r)}(\theta)\right)$$

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Time-average estimator

- Define the observed and simulated time-averages of the random features by

$$F^{\text{obs}} = \frac{1}{n-m} \sum_{t=m+1}^n f_t^{\text{obs}}, \quad \bar{F}^{\text{sim}}(\theta) = \frac{1}{n-m} \sum_{t=m+1}^n \bar{f}_t^{\text{sim}}(\theta)$$

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where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s} \sum_{r=1}^s f_t^{(r)}(\theta)$

- The time-average estimator is given by

$$\hat{\theta}^{\text{TA}} = \underset{\theta \in \Theta}{\operatorname{argmin}} \left\| \hat{Q}_n^{\text{TA}}(\theta) \right\|$$

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- In practice, an optimization procedure is used to find the minimizer of $\left\| \hat{Q}_n^{\text{TA}}(\cdot) \right\|$

Rolling-window estimator

- Define the observed and simulated rolling-window random features at time t by

$$F_t^{\text{obs}} = \frac{1}{w \wedge t} \sum_{j=(t-w) \vee 1}^t f_j^{\text{obs}}, \quad \bar{F}_t^{\text{sim}}(\theta) = \frac{1}{w \wedge t} \sum_{j=(t-w) \vee 1}^t \bar{f}_j^{\text{sim}}(\theta),$$

where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s} \sum_{r=1}^s f_t^{(r)}(\theta)$ and $w \in \mathbb{N}$ is the window size

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$$\hat{\theta}^{\text{RW}} = \underset{\theta \in \Theta}{\operatorname{argmin}} \left| \hat{Q}_n^{\text{RW}}(\theta) \right|,$$

where the sample discrepancy function is defined as

$$\hat{Q}_n^{\text{RW}}(\theta) = \frac{1}{n-m} \sum_{t=m+\tau+L}^n \left[\left\| F_t^{\text{obs}} - \bar{F}_t^{\text{sim}}(\theta) \right\|^2 + K_t(\theta) \right],$$
$$K_t(\theta) = 2 \left(F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta) \right)^{\top} \left(\left[f_t^{\text{obs}} - \bar{f}_t^{\text{sim}}(\theta) \right] - \left[F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta) \right] \right),$$

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Example: Mean of Gaussian

- **Goal:** Estimate $\theta_0 = \mu_0 = 0.5$ given $X_1^{\text{obs}}, \dots, X_n^{\text{obs}} \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta_0, 1)$ with $\Theta = [-3, 3]$

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- ▶ Use time-averages of $2p + 1 = 3$ random Fourier features with $m = 0$ lags
- ▶ For $i = 1, \dots, 2p + 1$, draw frequencies $w_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, phases $b_i \stackrel{\text{iid}}{\sim} \text{Unif}(-\pi, \pi)$

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- ▶ For $i = 1, \dots, 2p + 1$, draw frequencies $w_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, phases $b_i \stackrel{\text{iid}}{\sim} \text{Unif}(-\pi, \pi)$
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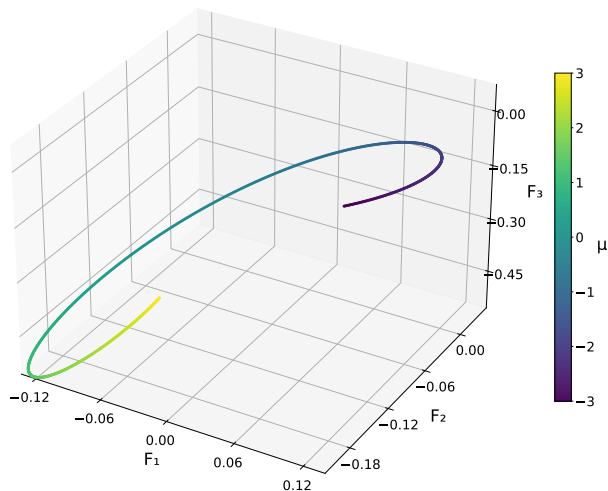
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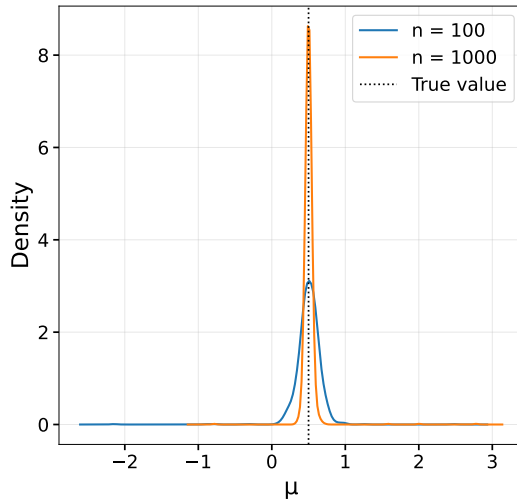
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- ▶ Let's visualize an embedding and check the estimation results

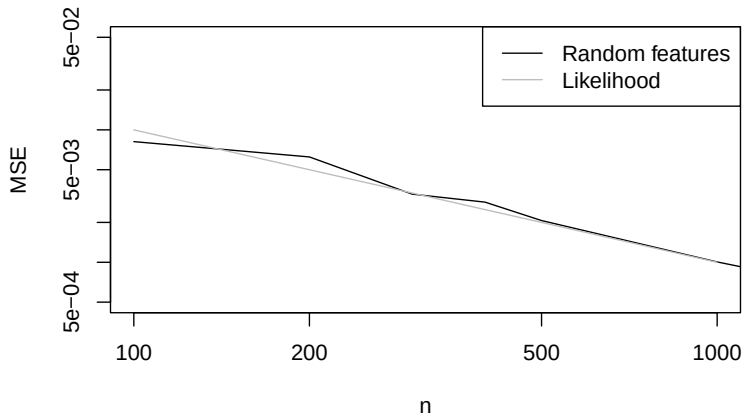
Each point is a time-average of 3 features, colored by value of parameter μ



Density plot from 1,000 independent estimates (different random features)



Theoretical MSE of $\hat{\theta}^{\text{MLE}}$ ($1/n$) vs MSE of our $\hat{\theta}$ (100 replicates per n)



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- ▶ **Key assumptions:** Decay of temporal dependence and control of nonstationarity

Representation in terms of noise inputs (Wu 2005)

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Assumption

Given a parameter $\theta \in \Theta$, sample size $n \in \mathbb{N}$, and noise inputs ε_t , a time series $X_{1:n}$ is generated as

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- For all $j \in \mathbb{N}$, denote $\varepsilon_t = (\varepsilon_t, \varepsilon_{t-1}, \dots)$ with ε_{t-j} replaced by an iid copy ε_{t-j}^* as

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Assumption

There exist constants $C > 0$, $\rho > 1$ such that, for some $q > 2$, we have

$$\sup_{n,t,\theta} \left\| G_t^{(n)}(\varepsilon_t, \theta) - G_t^{(n)}(\tilde{\varepsilon}_{t,j}, \theta) \right\|_{L^q(\theta)} \leq C j^{-\rho}, \quad j \in \mathbb{N}$$

where $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_{P_\theta} (\|\cdot\|_2^q)^{1/q}$

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- We measure the nonstationarity using a p -variation-type quantity, define

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Assumption

For some $q > 2$, some $p \in [1, 2)$, and all $i \in \mathbb{N}$, there exists a constant $\Lambda > 0$ such that

$$\sup_{\theta \in \Theta} \sup_{u \in [0,1]} \left\| \tilde{G}_u^{(i)}(\varepsilon_0, \theta) \right\|_{\mathcal{L}^q(\theta)} + \sup_{\theta \in \Theta} \left\| \left(\tilde{G}_u^{(i)}(\varepsilon_0, \theta) \right)_u \right\|_{p\text{-var}, \mathcal{L}^q(\theta)} \leq \Lambda$$

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- ▶ **Rosenblatt transform:** For each $j \in [d]$, sample $X_{t,j}$ from the conditional quantile given $X_{t,1}, \dots, X_{t,j-1}$ by transforming U_j (see Wu and Mielniczuk (2010))

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- ▶ Extensions to spatiotemporal data, network data, etc.
- ▶ In what other areas of statistics can random features be used?

Thank you!

Questions and comments are welcome
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Carnegie Mellon University

Definition of smooth manifold

Definition

A C^∞ -manifold (i.e. smooth manifold or differentiable manifold) is a Hausdorff topological space M , with countable basis, together with a C^∞ -atlas

- ▶ The next slides explain the definitions needed to understand this
- ▶ Source: Lecture notes from Eckhard Meinrenken at the University of Toronto
<https://www.math.utoronto.ca/mein/teaching/LectureNotes/Man.pdf>

Definition of smooth manifold

- ▶ An n -dimensional manifold is a space that locally looks like $\mathbb{R}^n$. To give a precise meaning to this idea, our space first of all has to come equipped with some topology (so that the word “local” makes sense)
- ▶ Recall that a *topological space* is a set M , together with a collection of subsets of M , called *open subsets*, satisfying the following three axioms: (i) the empty set $\emptyset$ and the space M itself are both open, (ii) the intersection of any finite collection of open subsets is open, (iii) the union of any collection of open subsets is open
- ▶ The collection of open subsets of M is also called the topology of M . A map $f : M_1 \rightarrow M_2$ between topological spaces is called *continuous* if the pre-image of any open subset in M_2 is open in M_1 . A continuous map with a continuous inverse is called a *homeomorphism*

Definition of smooth manifold

- ▶ One ingredient in the definition of a manifold is that our topological space comes equipped with a covering by open sets which are homeomorphic to open subsets of $\mathbb{R}^n$
- ▶ Let M be a topological space. An n -dimensional *chart* for M is a pair (U, ϕ) consisting of an open subset $U \subset M$ and a continuous map $\phi : U \rightarrow \mathbb{R}^n$ such that ϕ is a homeomorphism onto its image $\phi(U)$. Two such charts (U_α, ϕ_α) , (U_β, ϕ_β) are C^∞ -compatible if the transition map

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$$

is a diffeomorphism (a smooth map with smooth inverse). A covering $\mathcal{A} = (U_\alpha)_{\alpha \in A}$ of M by pairwise C^∞ -compatible charts is called a C^∞ -atlas

Definition of smooth manifold

- ▶ We recall that a topological space is called *Hausdorff* if any two points have disjoint open neighborhoods.
- ▶ A *basis* for a topological space M is a collection $\mathcal{B}$ of open subsets of M such that every open subset of M is a union of open subsets in the collection $\mathcal{B}$.
- ▶ A topological space with countable basis is also called *second countable*
- ▶ For example, the collection of open balls $B_\varepsilon(x)$ in $\mathbb{R}^n$ define a basis. But one already has a basis if one takes only all balls $B_\varepsilon(x)$ with $x \in \mathbb{Q}^n$ and $\varepsilon \in \mathbb{Q}_{>0}$; this defines a countable basis.

References I

-  Kallenberg, Olav (2021). *Foundations of modern probability*. Third edition. Springer.
-  Sauer, Tim, James A. Yorke, and Martin Casdagli (1991). “Embedology”. In: *Journal of Statistical Physics* 65.3, pp. 579–616.
-  Wu, Wei Biao (2005). “Nonlinear system theory: another look at dependence”. In: *Proceedings of the National Academy of Sciences* 102.40, pp. 14150–14154.
-  Wu, Wei Biao and Jan Mielniczuk (2010). “A new look at measuring dependence”. In: *Dependence in Probability and Statistics*, pp. 123–142.
-  Yorke, James and William Ott (2005). “Prevalence”. In: *Bulletin of the American Mathematical Society* 42.3, pp. 263–290.